

CMPE 482 – Multimedia Networks

Homework 1: Video Coding

Experiments with FFmpeg

Qatar University
Computer Science and Engineering Department
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Group Members:

- your name (QUID: XXXXXXXX)
- <Add others if group work>

1. Introduction

The aim of this assignment is to analyze the effect of quantization parameter (QP), GOP length, and B-frames on bitrate, PSNR, and SSIM. Additionally, we compare the efficiency of H.264 (libx264) and H.265 (libx265) codecs. The analysis is based on both objective metrics and subjective visual assessment.

2. Methodology

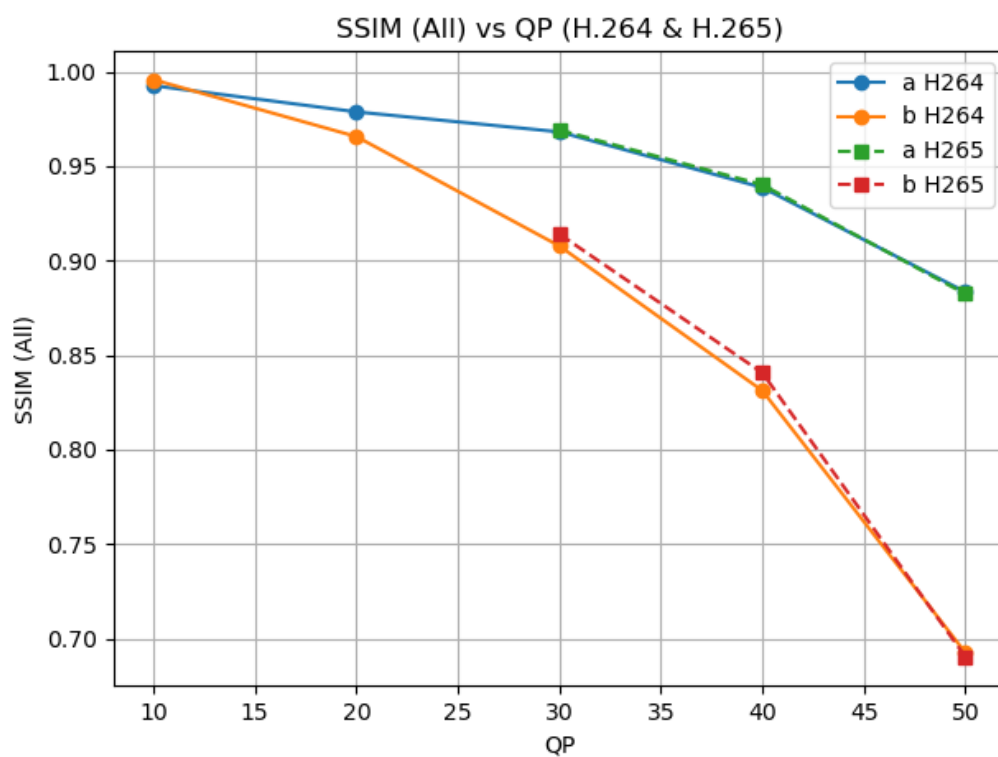
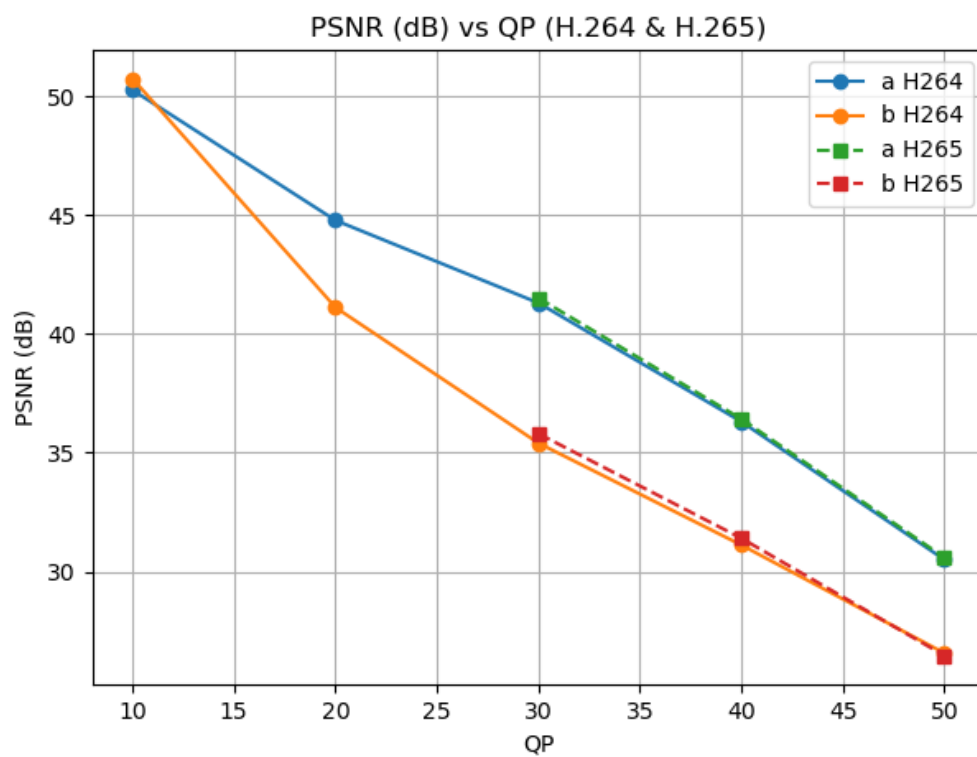
Experiments were conducted using Ubuntu WSL on Windows 11 with VSCode terminal.

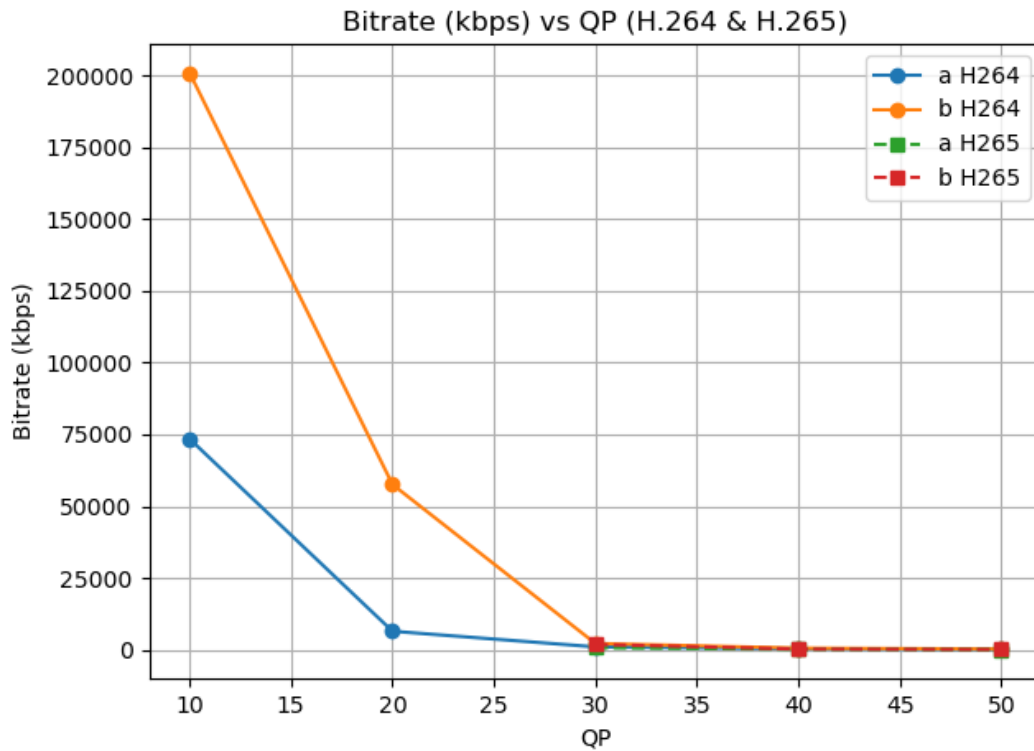
- FFmpeg version: 6.1.1-3ubuntu5
- Input videos: a.y4m, b.y4m, noise.mp4
- Scripts: Automated shell scripts were used to encode videos and measure quality.
- Bitrate formula: $\text{bitrate (kbps)} = (\text{file_size} * 8) / (\text{duration} * 1000)$
- Metrics: PSNR and SSIM computed with FFmpeg filters.
- Figures generated with Python + Matplotlib.

3. Results and Discussion

3.1 Effect of QP (H.264)

Figures: Figure 1.1 (PSNR vs QP), Figure 1.2 (SSIM vs QP), Figure 1.3 (Bitrate vs QP).





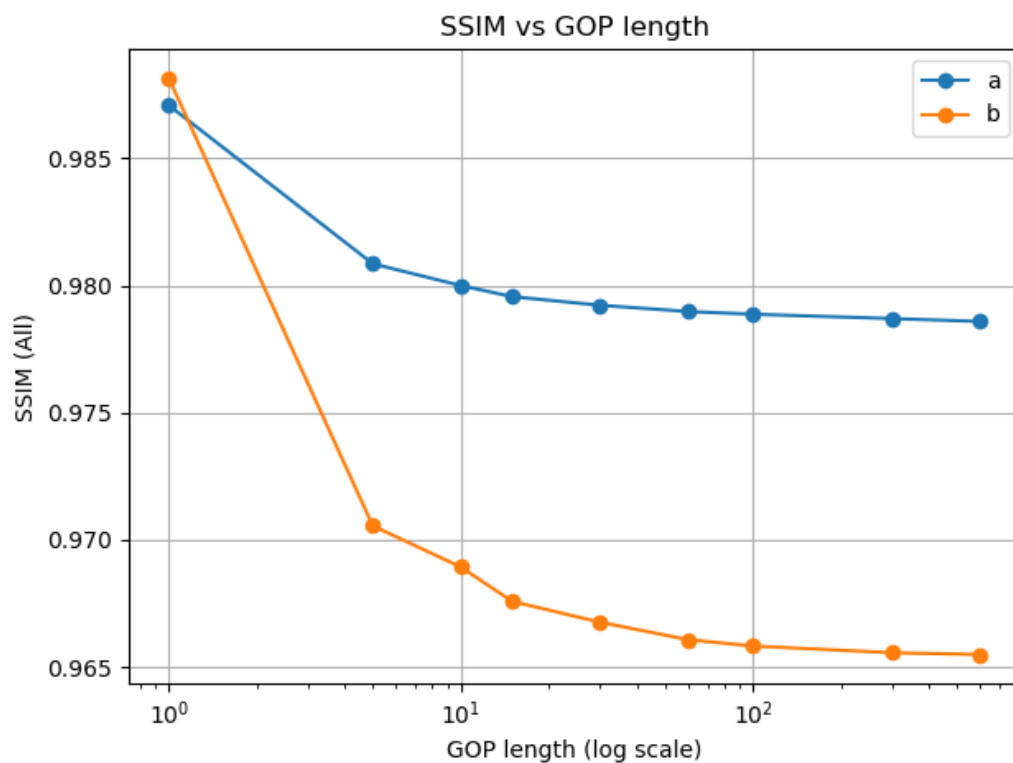
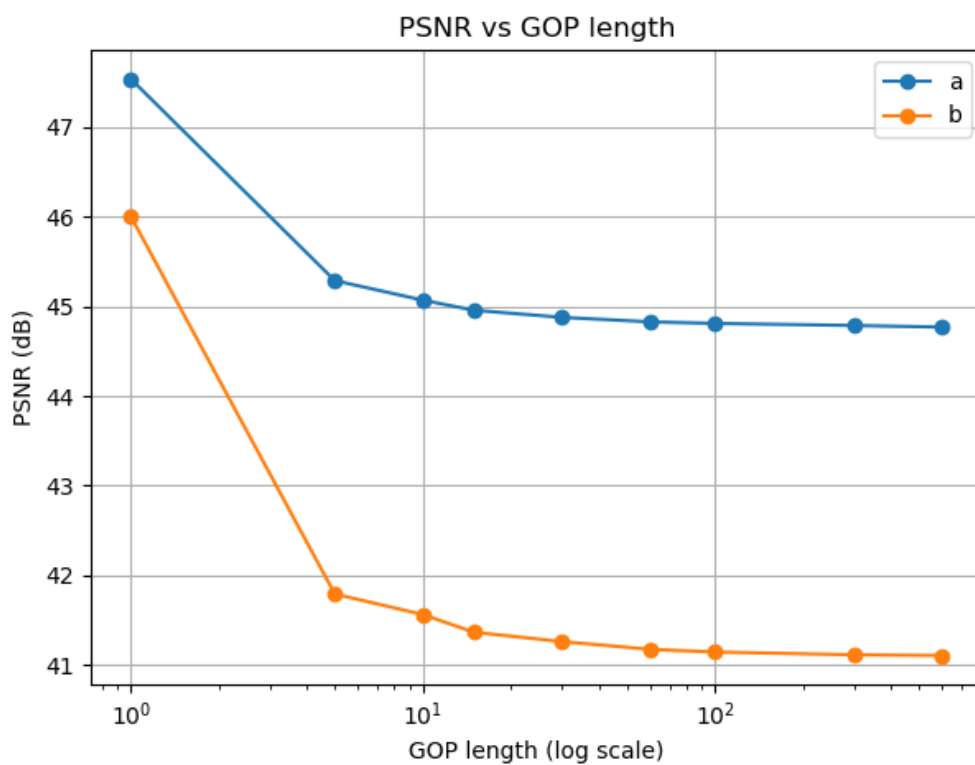
For a.y4m, bitrate dropped from **~73441 kbps** at QP=10 to **~120 kbps** at QP=50. PSNR decreased from **~44 dB** to **~30 dB**. SSIM stayed near **0.99** at low QP and dropped below **0.94** at high QP.

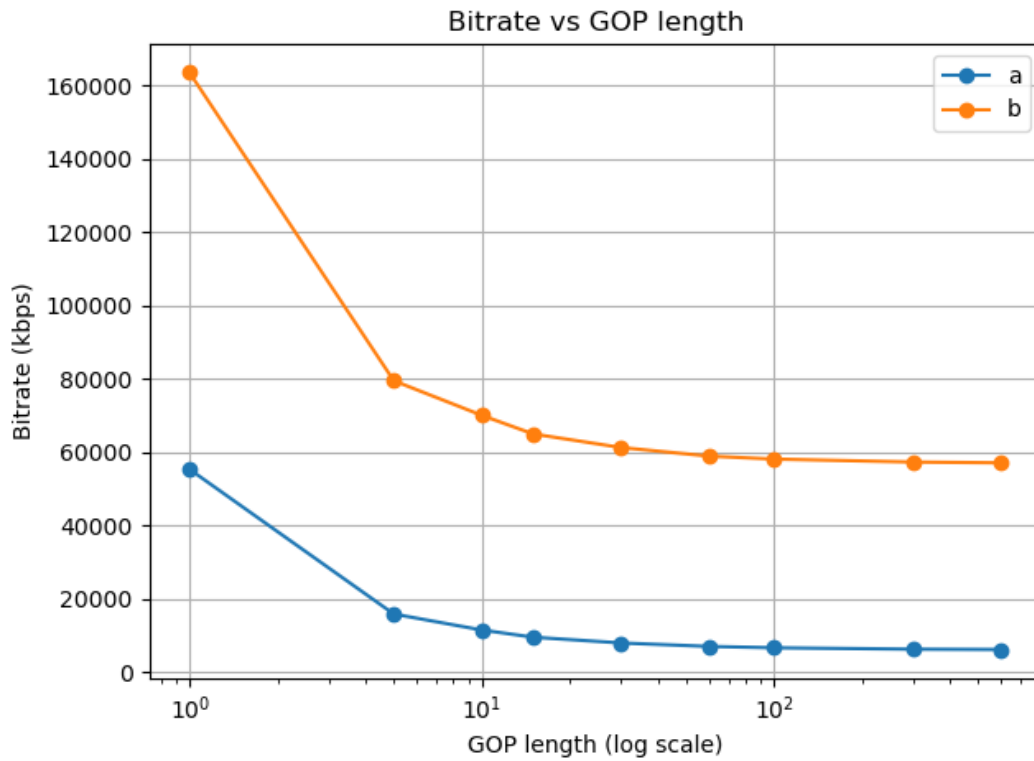
For b.y4m, similar trends were observed, with bitrate from **~200975 kbps** to **~253 kbps**, and PSNR from **~50 dB** to **~26 dB**.

Observation: a.y4m showed visible degradation earlier (around QP=40). Noise video produced low PSNR (~10–12 dB) and SSIM (~0.01–0.05), showing SSIM better matches human perception.

3.2 Effect of GOP Length (QP=20)

Figures: Figure 2.1 (PSNR vs GOP), Figure 2.2 (SSIM vs GOP), Figure 2.3 (Bitrate vs GOP).



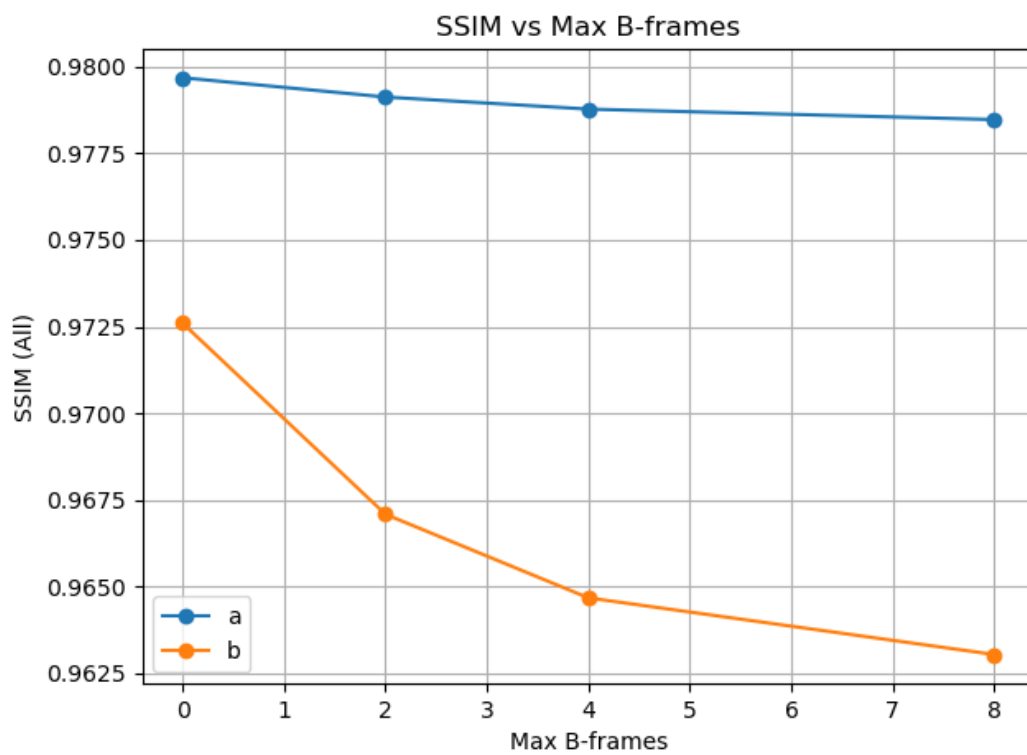
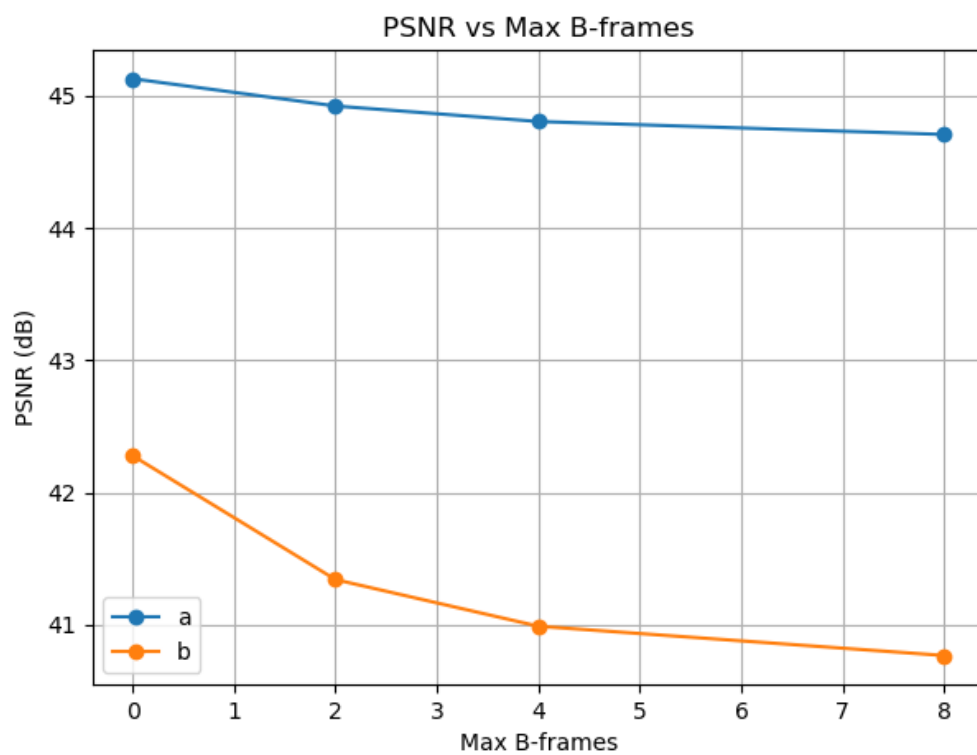


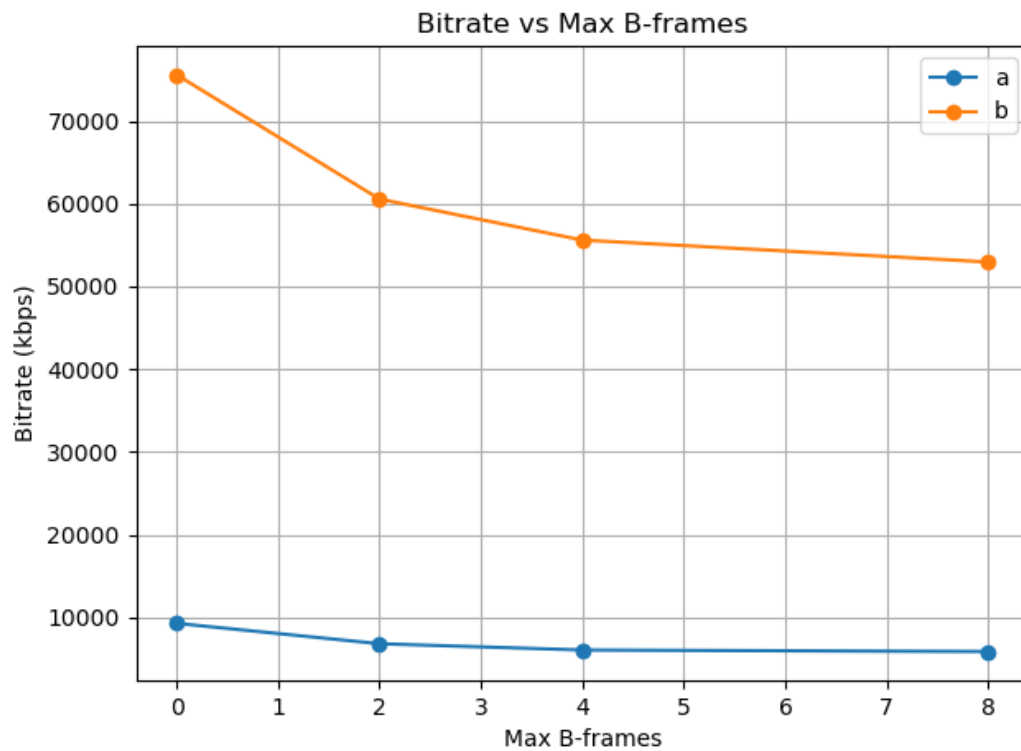
At GOP=1 (all intra), bitrate was highest (~55336 kbps for a.y4m, ~163380 kbps for b.y4m). As GOP increased to 100–600, bitrate dropped to ~6239 kbps for a.y4m and ~57107kbps for b.y4m while PSNR remained ~47–44 dB for a.y4m and ~46–41 dB for b.y4m and SSIM >0.95.

Observation: Longer GOP reduces bitrate significantly without hurting quality, but increases error propagation risk.

3.3 Effect of B-frames (QP=20, GOP=250)

Figures: Figure 3.1 (PSNR vs B-frames), Figure 3.2 (SSIM vs B-frames), Figure 3.3 (Bitrate vs B-frames).



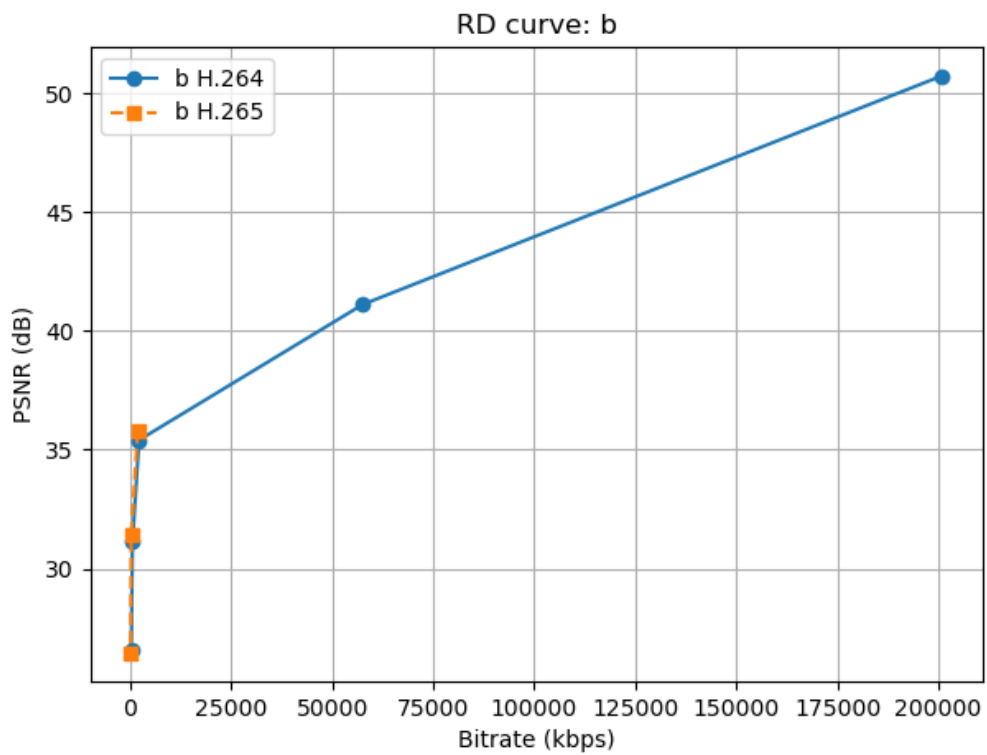
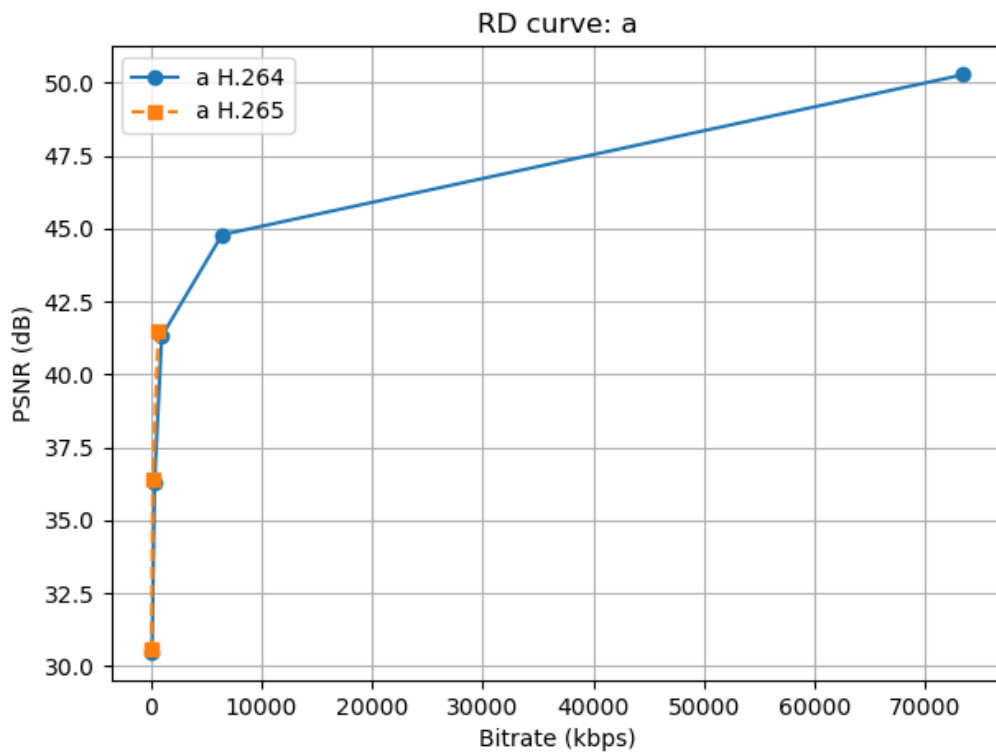


At 0 B-frames, bitrate was ~9280 kbps (a.y4m). With 8 B-frames, bitrate dropped to ~5875 kbps, while PSNR improved slightly (~+1 dB) and SSIM stayed >0.97.

Observation: B-frames improve coding efficiency but introduce latency, unsuitable for real-time apps.

3.4 Codec Comparison (H.264 vs H.265)

Figures: Figure 4.1–4.2 (PSNR/SSIM/Bitrate vs QP, RD curves).



Example (a.y4m, QP=40): H.264 bitrate ~345 kbps, PSNR ~36 dB. H.265 bitrate ~186 kbps, PSNR ~36 dB. Thus, H.265 achieves ~54% bitrate savings.

Observation: H.265 consistently outperformed H.264 in compression efficiency but required longer encoding times.

4. Conclusion

- Lower QP → higher quality but higher bitrate.
- Longer GOP → bitrate savings with stable quality, but less error resilience.
- B-frames → bitrate savings with small delay cost.
- H.265 → up to 54% bitrate reduction compared to H.264.
- SSIM correlates better with human perception than PSNR, especially with noise video.